

Thème 5 — Trigonométrie, exponentielle & logarithme

Corrigé — niveau Moyen

*Détail des calculs.***Exercice 1**

$$(a) 2^x = 8 = 2^3 \Rightarrow \boxed{x = 3}. \quad (b) e^x = 1 = e^0 \Rightarrow \boxed{x = 0}. \quad (c) 3^x = \frac{1}{9} = 3^{-2} \Rightarrow \boxed{x = -2}.$$

Exercice 2

$$(a) \ln x = 2 \Rightarrow x = e^2 \text{ (et } e^2 > 0, \text{ cohérent) : } \boxed{x = e^2}.$$
$$(b) \ln(x - 1) = 0 \Rightarrow x - 1 = e^0 = 1 \Rightarrow x = 2 \text{ (et } x - 1 = 1 > 0, \text{ cohérent) : } \boxed{x = 2}.$$

Exercice 3

$$\frac{13\pi}{3} = \frac{12\pi}{3} + \frac{\pi}{3} = 4\pi + \frac{\pi}{3}. \text{ Comme } \cos \text{ est } 2\pi\text{-périodique, on retranche 2 tours } (4\pi) :$$

$$\cos \frac{13\pi}{3} = \cos \left(\frac{\pi}{3} \right) = \boxed{\frac{1}{2}}.$$

Exercice 4

$$\arcsin 1 = \frac{\pi}{2} \text{ (car } \sin \frac{\pi}{2} = 1); \arccos 0 = \frac{\pi}{2} \text{ (car } \cos \frac{\pi}{2} = 0); \arctan 1 = \frac{\pi}{4} \text{ (car } \tan \frac{\pi}{4} = 1).$$

Exercice 5

$$a) \text{ Changement de base : } \log_2 5 = \frac{\ln 5}{\ln 2}.$$
$$b) e^{2x} = e^{x+1}. \text{ L'exponentielle étant injective (strictement croissante) : } 2x = x + 1 \Rightarrow \boxed{x = 1}.$$